WealthWise AI

AI-Powered Personal Finance Advisor

Project Report

June 2026

1. Project Overview

1.1 Problem Statement

Managing personal finances is a challenge for most individuals. Bank statements contain large volumes of transaction data that are difficult to interpret manually. Users struggle to understand their spending habits, estimate savings rates, assess financial risk, and identify suitable investment options — all without professional guidance.

1.2 Proposed Solution

WealthWise AI is an AI-powered web application that allows users to upload their bank statements (PDF or CSV). The system automatically extracts transactions, analyzes spending patterns, classifies the user's financial risk profile using Machine Learning, generates a personalized investment portfolio, and delivers natural-language financial advice powered by the Gemini API — all through an interactive dashboard.

1.3 Target Users

- Salaried professionals seeking clarity on personal spending and savings.
- College students managing independent finances for the first time.
- First-time investors looking for data-driven portfolio guidance.

1.4 Project Scope

In Scope:

- Bank statement upload, transaction extraction, categorization, and analytics.
- ML-based financial risk profile classification.
- Portfolio recommendation and AI-generated financial advice.
- Financial Health Score computation.
- Cloud deployment on AWS with Docker.

Out of Scope:

- Real-time stock prediction or live trading.
- Loan approval or credit scoring.
- Multi-bank live API integrations.
- Mobile application (iOS / Android).

1.5 Requirement Gathering

Functional Requirements

- User registration and login.
- Upload PDF or CSV bank statements.
- Extract transactions from uploaded statements.
- Categorize expenses into meaningful groups.
- Generate income, expense, and savings analytics.
- Predict the user's financial risk profile (Conservative / Moderate / Aggressive).
- Recommend a portfolio allocation based on the risk profile.
- Generate AI-based financial advice.
- View historical financial reports.

Non-Functional Requirements

- Secure authentication for all user accounts.
- Fast statement processing with minimal delay.
- User-friendly and intuitive dashboard.
- Accurate transaction extraction from bank statements.
- Scalable database design to support growing data.
- Cloud-deployable system architecture.
- Data privacy and security for all user information.

1.6 Types of Users & Features

WealthWise AI has two distinct user types — Regular User and Admin. Each has a defined role, set of features, and a complete journey from registration to logout.

User Types Overview

User Type	Who Is This?	Access Level
Regular User	Salaried individual, student, or first-time investor who uploads bank statements for personal finance analysis.	Personal dashboard, statement upload, analytics, AI advice.
Admin User	Platform operator / developer who monitors system health and manages user accounts.	User management, activity logs, system monitoring.

Regular User — Complete Journey & Features

#	Step	What Happens / Feature
1	Landing Page	Visits the website. Sees platform overview and 'Get Started' button.
2	Register	Fills name, email, mobile, password, occupation. Email OTP verification required.
3	Login / Forgot Password	Email + password login. JWT token issued. OTP-based password reset available.
4	Dashboard (Home)	Sees Financial Health Score, income/expense summary, and Smart Alerts at a glance.
5	Upload Bank Statement	Drag-and-drop PDF or CSV (max 10 MB). System validates, stores in S3, begins processing.
6	View Transactions	Clean table of all extracted transactions: Date, Description, Debit, Credit, Category.
7	Expense Analytics	Pie chart (spending by category), bar chart (income vs expenses), savings rate displayed.
8	Financial Health Score	Score 0-100 based on savings rate, expense stability, and income consistency.
9	Risk Profile (ML)	ML model predicts: Conservative / Moderate / Aggressive with explanation.
10	Portfolio Recommendation	Personalized asset allocation chart: Stocks, Mutual Funds, Gold, FD, Crypto.
11	AI Financial Coach	Gemini API generates a personalized natural-language advisory paragraph.
12	Smart Alerts	Flags shown when any spending category exceeds its historical average.
13	Historical Reports	View past monthly analyses, compare months, download PDF summary reports.
14	Profile & Settings	Update name, change password, manage preferences, or delete account.

15	Logout	JWT token cleared. Session ends. Redirected to landing page.

Admin User — Complete Journey & Features

#	Step	What Happens / Feature
1	Admin Login	Logs in with admin credentials. Role-based JWT token with admin permissions issued.
2	Admin Dashboard	Sees total users, total uploads, processing success/failure stats, and system health.
3	User Management	View all registered users. Activate, deactivate, or delete any user account.
4	Upload Activity Log	View every file uploaded across all users: file name, timestamp, status (processed / failed).
5	System Monitoring	AWS S3 usage, API response times, failed extraction jobs, Gemini API call count.
6	Logout	Admin session terminated. Token invalidated. Redirected to login page.

2. Existing Systems & Gap Analysis

2.1 Existing Platforms Studied

Before designing WealthWise AI, we studied the following existing personal finance platforms available in India:

Platform	What It Provides
Groww	Investment options, mutual fund investments, portfolio tracking, and financial products for retail investors.
ET Money	Mutual fund investments, SIP management, expense tracking, and basic financial products.
INDmoney	Wealth tracking, US stocks, mutual funds, EPF/PPF tracking, and net worth dashboard.

2.2 Limitations of Existing Platforms

Platform	Limitation
Groww	Focuses mainly on investing; does not analyze uploaded bank statements or understand spending behavior.
ET Money	Provides financial products but limited AI-based spending analysis; no statement upload or risk prediction.
INDmoney	Tracks overall wealth but does not generate detailed AI-driven insights from bank statements.
WealthWise AI	**Combines statement analysis, ML risk prediction, portfolio recommendation, and AI financial coaching in one unified platform.**

2.3 What Makes WealthWise AI Unique

Most existing platforms focus on showing investment options and tracking portfolios. WealthWise AI goes significantly further:

What Most Apps Do	What WealthWise AI Does
Show investment options	**Understands actual spending behavior from bank statements**
Track existing portfolio	**Analyzes bank statements automatically (PDF & CSV)**

What Most Apps Do	What WealthWise AI Does
Suggest generic financial products	**Predicts risk profile using a trained ML model**
Require manual data entry	**Generates personalized investment allocation automatically**
No AI-driven personalization	**Provides AI-generated financial advice via Gemini API**
No financial health assessment	**Calculates a Financial Health Score based on real data**

3. System Features

Feature	Description
User Registration & Login	Secure sign-up, email verification, JWT login, logout, and password reset via OTP.
Statement Upload	PDF and CSV upload with drag-and-drop, file validation, and upload history tracking.
Transaction Extraction	Automated extraction of date, description, debit, credit, and balance from PDFs (pdfplumber + pytesseract OCR) and CSV files.
Transaction Categorization	Keyword-based mapping to categories: Food, Shopping, Bills, Travel, Healthcare, Entertainment, Investment, Salary, Others.
Financial Analytics Dashboard	Monthly income/expense charts, category-wise spending breakdown, savings rate, and balance trends.
Financial Health Score	A composite score calculated from savings rate, expense stability, and income consistency, giving users a clear measure of financial wellness.
ML Risk Classification	Random Forest model classifying users as Conservative, Moderate, or Aggressive based on financial behavior.
Portfolio Recommendation	Personalized asset allocation (Stocks, Mutual Funds, Gold, FD, Crypto) based on the predicted risk profile.
AI Financial Coach	Gemini API generates a personalized natural-language financial summary with actionable recommendations.
Smart Alerts	Automated flags when any spending category exceeds historical averages or defined thresholds.
Historical Reports	Month-wise financial summaries, downloadable analytics reports, and statement history.
Security	bcrypt password hashing, JWT tokens (HTTP-only cookies), HTTPS, and private AWS S3 storage.

4. Application User Flow

The following flow describes the end-to-end user journey through the WealthWise AI platform, from initial landing to final report generation.

1. Landing Page

▼

2. Register / Login

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3. Dashboard

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4. Upload Bank Statement

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5. Transaction Extraction

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6. Expense Categorization

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7. Financial Analytics

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8. Financial Health Score

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9. ML Risk Prediction

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10. Portfolio Recommendation

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11. AI Financial Coach

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12. Smart Alerts

▼

13. Historical Reports

5. Technology Stack

Category	Technology / Tools
Frontend	React.js, Tailwind CSS, Chart.js
Backend	FastAPI (Python)
Database	PostgreSQL
Data Processing	Pandas, NumPy
PDF Extraction	pdfplumber, pytesseract (OCR)
Machine Learning	Scikit-learn — Random Forest Classifier
Generative AI	Gemini API (Google)
Cloud	AWS EC2, AWS S3
DevOps	Docker

6. System Architecture

WealthWise AI uses a layered architecture that separates the frontend, API, business logic, ML, GenAI, and data storage concerns cleanly. The end-to-end data flow is as follows:

1. User authenticates and uploads a bank statement via the React.js frontend.
2. The FastAPI backend validates the file and stores it in AWS S3.
3. The extraction pipeline (pdfplumber / pytesseract / Pandas) parses the file and stores transactions in PostgreSQL.
4. The Analytics Engine computes income, expenses, savings rate, and category breakdowns.
5. The Financial Health Score is calculated from the analytics metrics.
6. The Random Forest Classifier predicts the user's financial risk profile.
7. The Portfolio Recommendation Engine generates asset allocation based on the risk profile.
8. The Gemini API synthesizes all data into a personalized natural-language advisory.
9. Results are returned to the React dashboard for interactive visualization.

Layer	Component	Technology
Presentation	User Interface & Dashboard	React.js, Tailwind CSS, Chart.js
API Gateway	REST API Endpoints	FastAPI (Python)
Business Logic	Financial Analytics Engine	Python, Pandas, NumPy

Layer	Component	Technology
ML Pipeline	Risk Classification Model	Scikit-learn, Random Forest
GenAI Layer	AI Financial Coach	Gemini API
Data Layer	Relational Database	PostgreSQL
File Storage	Uploaded Statement Files	AWS S3
Compute	Application Server	AWS EC2, Docker

7. Functional Modules

Module 1 — User Authentication

- Registration with email verification, JWT login, logout, and OTP-based password reset.
- Passwords hashed with bcrypt; tokens stored in HTTP-only cookies.

Module 2 — Statement Upload

- Supports PDF and CSV bank statements (max 10 MB).
- Files stored in private AWS S3; upload record and processing status tracked in PostgreSQL.
- Duplicate detection using SHA-256 file hashing to prevent re-processing.

Module 3 — Transaction Extraction

- PDF: pdfplumber for machine-readable PDFs; pytesseract OCR for scanned PDFs.
- CSV: Pandas-based parsing with automatic column mapping.
- Fields extracted: Transaction Date, Description, Debit, Credit, Balance.

Module 4 — Transaction Categorization

Transactions are classified using keyword mapping into the following categories:

Category	Example Keywords / Merchants
Food & Dining	Swiggy, Zomato, McDonald's, BigBasket, Blinkit
Shopping	Amazon, Flipkart, Myntra, Nykaa, AJIO
Travel & Transport	Uber, Ola, MakeMyTrip, IRCTC, IndiGo
Bills & Utilities	Electricity, BSNL, Airtel, Jio, Water Bill
Healthcare	Apollo, Medplus, PharmEasy, hospital, clinic
Entertainment	Netflix, Spotify, BookMyShow, PVR, Amazon Prime
Investment	Zerodha, Groww, SIP, mutual fund, NSE, BSE
Salary / Income	SALARY, NEFT CR, Opening Balance
Others	All unmatched transactions

Module 5 — Financial Analytics & Health Score

- Monthly Income, Expenses, Savings, and Savings Rate (%).

- Category-wise spending breakdown and month-over-month expense trends.
- Financial Health Score: A composite metric derived from savings rate, expense stability, and income consistency — gives users a single, easy-to-understand measure of financial wellness.

Module 6 — ML Risk Classification

- Algorithm: Random Forest Classifier (Scikit-learn).
- Input Features: Monthly Income, Monthly Expenses, Savings Rate, Average Account Balance.
- Output Classes: Conservative | Moderate | Aggressive.

Module 7 — Portfolio Recommendation

Asset Class	Conservative	Moderate	Aggressive
Stocks / Equity	10%	30%	50%
Mutual Funds / SIP	30%	40%	30%
Gold / Sovereign Gold Bond	20%	15%	10%
Fixed Deposit (FD)	40%	15%	0%
Crypto / Alt Assets	0%	0%	10%

Module 8 — AI Financial Coach

- Input: Financial analytics, risk profile, and portfolio recommendation.
- Output: Personalized natural-language financial summary via Gemini API.

Sample output: "Your savings rate of 24% is above the recommended 20% benchmark. Food & Dining represents 31% of monthly spend — higher than average. Based on your Moderate risk profile, increasing your SIP allocation by 10% could significantly improve long-term wealth accumulation."

8. Dataset

Dataset: AgamiAI Indian Bank Statements — Hugging Face Repository.

- Components: Digital PDFs, Scanned PDFs, and JSON Metadata.
- Transaction fields: Date, Description, Debit, Credit, Account Balance.
- Usage: PDF extraction testing, transaction parsing, analytics computation, and ML model training for risk classification.

9. Database Design

Table: Users

Field	Data Type	Description
user_id	UUID	PK — Unique user identifier.
name	VARCHAR(100)	Full name.
email	VARCHAR(150)	Unique email for login.
password_hash	VARCHAR(255)	bcrypt-hashed password.
is_verified	BOOLEAN	Email verification status.

Table: Uploaded_Statements

Field	Data Type	Description
statement_id	UUID	PK.
user_id	FK → Users	Owner of the statement.
file_name	VARCHAR(255)	Original file name.
s3_key	VARCHAR(500)	AWS S3 storage key.
status	VARCHAR(50)	pending / processed / failed.

Table: Transactions

Field	Data Type	Description
transaction_id	UUID	PK.
statement_id	FK → Statements	Linked statement.
date	DATE	Transaction date.
description	TEXT	Merchant / narration.
debit	NUMERIC(12,2)	Outflow amount.
credit	NUMERIC(12,2)	Inflow amount.
balance	NUMERIC(12,2)	Closing balance.
category	VARCHAR(50)	Assigned category.

Table: Analytics

Field	Data Type	Description
analytics_id	UUID	PK.
user_id	FK → Users	Linked user.
month_year	VARCHAR(10)	Period (e.g., 2025-05).
income	NUMERIC(12,2)	Total monthly income.
expenses	NUMERIC(12,2)	Total monthly expenses.
savings_rate	NUMERIC(5,2)	Savings as % of income.
health_score	NUMERIC(5,2)	Financial Health Score.

Table: Risk_Profile

Field	Data Type	Description
risk_id	UUID	PK.
user_id	FK → Users	Linked user.
risk_level	VARCHAR(20)	Conservative / Moderate / Aggressive.

Table: Portfolio_Recommendation

Field	Data Type	Description
recommendation_id	UUID	PK.
user_id	FK → Users	Linked user.
stocks_pct	NUMERIC(5,2)	Stocks allocation.
mutual_funds_pct	NUMERIC(5,2)	Mutual Funds.
gold_pct	NUMERIC(5,2)	Gold.
fd_pct	NUMERIC(5,2)	Fixed Deposit.
crypto_pct	NUMERIC(5,2)	Crypto / Alt Assets.

10. Agile Development Lifecycle

WealthWise AI follows the Agile Scrum methodology. Given a 45-day internship duration, development is structured into four focused sprints, each delivering a testable and deployable increment.

10.1 Agile Roles

- Product Owner: Maintains and prioritizes the product backlog; defines acceptance criteria.
- Scrum Master: Facilitates sprint ceremonies and removes blockers.
- Development Team: Handles design, development, testing, and deployment.

10.2 Sprint Ceremonies

- Sprint Planning — Define goals and assign tasks at the start of each sprint.
- Daily Stand-up — Brief daily sync to share progress and surface blockers.
- Sprint Review — Demonstrate the working increment at sprint end.
- Sprint Retrospective — Reflect on improvements for the next sprint.

10.3 Sprint Plan (45-Day Schedule)

Sprint	Timeline	Theme	Key Tasks	Deliverable
Sprint 1	Days 1–10	**Foundation & Auth**	Project setup, DB schema design, user auth (register/login/JWT), AWS S3 configuration, Docker environment.	Working auth system, DB live, S3 integration complete.
Sprint 2	Days 11–22	**Core Pipeline**	Statement upload UI, pdfplumber extraction, pytesseract OCR, CSV parser, keyword-based categorization, analytics engine (income/expenses/savings rate).	End-to-end: upload → extract → categorize → analytics computed.
Sprint 3	Days 23–34	**ML, Portfolio & AI Advisor**	Feature engineering, Random Forest model training & evaluation, portfolio recommendation engine, Gemini API integration, Financial Health Score.	Risk classifier live; AI Advisor generating personalized summaries.
Sprint 4	Days 35–45	**Dashboard, Testing & Deploy**	React dashboard (Chart.js charts), Smart Alerts, Historical Reports, integration testing,	Live production deployment; fully

Sprint	Timeline	Theme	Key Tasks	Deliverable
			security hardening, AWS EC2 deployment.	tested interactive dashboard.

10.4 Definition of Done

- Code reviewed and merged to the main branch.
- Unit tests passing with >80% code coverage.
- Feature verified on the staging environment.
- API endpoints documented in FastAPI Swagger UI.
- All sprint acceptance criteria are met.

11. Success Criteria

#	Success Criterion
1	User can register, login, and authenticate securely.
2	Bank statements (PDF and CSV) upload and process successfully.
3	Transactions are accurately extracted from uploaded documents.
4	Transactions are correctly categorized into expense groups.
5	Financial analytics (income, expenses, savings rate) are computed correctly.
6	Financial Health Score is generated and displayed on the dashboard.
7	ML model predicts risk profile with accuracy > 80%.
8	Portfolio allocation is generated based on the predicted risk profile.
9	AI Financial Coach generates a coherent, personalized financial summary.
10	Application is deployed on AWS EC2 and accessible via a public URL.